Predict-Energy-Efficiency – AI101 Project

GitHub Repository Link



Damera Vyshali

Documentation Lead

2026EEB1338



Sahittya Dutta

Execution Lead

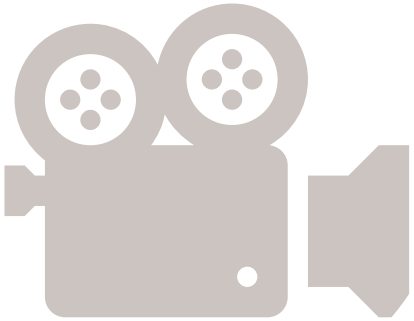
2026ESB1491



Haral Anuradha Dnyaneshwar

PPT Lead

2026AIB1039



Prathmesh Jamdade

Video Lead

2026MCB1555

Group 31

Problem Statement

What?

Predict a building's heating load and cooling load using its dimensions, glazing area, orientation, and other design characteristics.

Why?

Building design choices affect the energy needed to heat and cool indoor spaces. Estimating these loads early can help compare design options and identify choices associated with higher or lower demand before a building is constructed.



The Dataset

The Dataset Link:
https://archive.ics.uci.edu/dataset/242/energy+efficiency

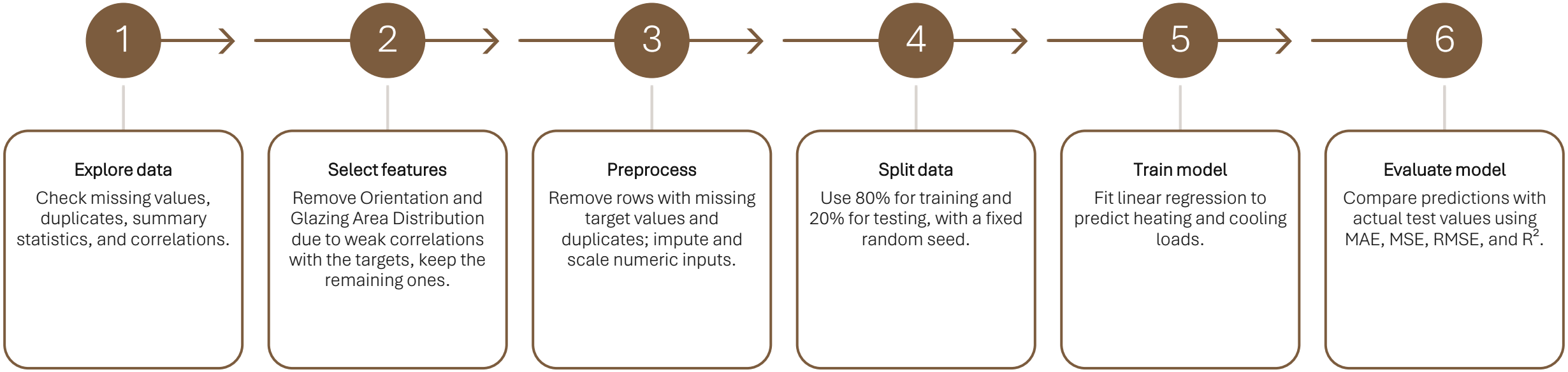
Instances (Rows): 768

10 Columns:
8 Input Columns and 2 Targets (Heating and Cooling Load)

Symbol	Variable	Unit
X1	Relative Compactness	Dimensionless
X2	Surface Area	m^2
X3	Wall Area	m^2
X4	Roof Area	m^2
X5	Overall Height	m
X6	Orientation	Dimensionless
X7	Glazing Area	Dimensionless
X8	Glazing Area Distribution	Dimensionless
Y1	Heating Load	kWh/m^2
Y2	Cooling Load	kWh/m^2

Approach

Pipeline

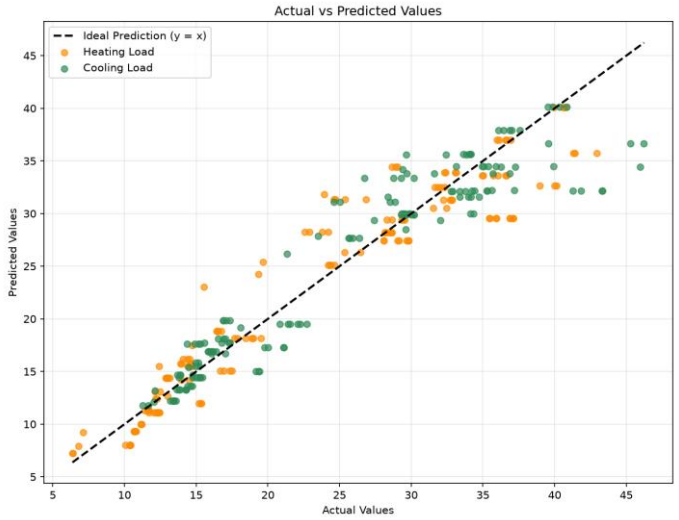


Libraries used

`contextlib`	`os`	`numpy`	`pandas`	`matplotlib.pyplot`	`scikit-learn`
Saves exploration output to a text file	Creates output folders and file paths	Supports numeric calculations such as RMSE	Reads and organizes the dataset	Creates and saves graphs	Provides Machine Learning tools

Results

Test Set: 154 Records (20% of the total dataset)



actual_vs_predicted.png

Target	MAE	MSE	RMSE	R2
Heating Load	2.164	9.218	3.036	0.912
Cooling Load	2.201	9.954	3.155	0.893

Different evaluation metrics on test data

```
(.venv) PS C:\Users\javad\Downloads\testing\Predict-Energy-Efficiency> python main.py
Dataset has been loaded sucessfully.
Dataset dimensions: (768, 10)
Column names:
['Relative Compactness', 'Surface Area', 'Wall Area', 'Roof Area', 'Overall Height', 'Orientation', 'Glazing Area', 'Glazing Are
ad', 'Cooling Load']
Rows with empty values found: 0. Rows with duplicate values found: 0.

The detailed Data Exploration has been saved as outputs/data_exploration_and_analysis/data_exploration_output.txt
The detailed Correlation Matrix has been saved as outputs/data_exploration_and_analysis/correlation_matrix.csv
The features vs targets plots have been saved as outputs/data_exploration_and_analysis/features_vs_targets.png


Feature selection:
Features: ['Relative Compactness', 'Surface Area', 'Wall Area', 'Roof Area', 'Overall Height', 'Glazing Area']
Targets: ['Heating Load', 'Cooling Load']

Model training complete.
The test predictions have been saved as outputs/prediction_results/test_predictions.csv

Testing Results for Heating Load:
MAE  : 2.164
MSE  : 9.218
RMSE : 3.036
R²   : 0.912

Testing Results for Cooling Load:
MAE  : 2.201
MSE  : 9.954
RMSE : 3.155
R²   : 0.893

The evaluation of the model using test data has been saved as outputs/prediction_results/evaluation_metrics.txt
The actual value vs predicted value plot has been saved as outputs/prediction_results/actual_vs_predicted.png
(.venv) PS C:\Users\javad\Downloads\testing\Predict-Energy-Efficiency>
(.venv) PS C:\Users\javad\Downloads\testing\Predict-Energy-Efficiency>
(.venv) PS C:\Users\javad\Downloads\testing\Predict-Energy-Efficiency>
```

Screenshot of the output when code is ran in the terminal

Observations

1. The model was slightly better on predicting the heating load compared to the cooling load
2. The model predicts the values pretty well (explains about 90% of variation for both targets)

Things that we got right

- Checked the data before modelling
- We used a pipeline so scaling is learned from the training data.

Possible Improvements

- Test the model in different geographical regions and varying climates
- Try a non-linear regression model

Takeways

- Made a model that predicts reasonably well
- Building geometry has a strong correlation with heating and cooling loads

Contributions

Component lead	Team member	Contribution
Documentation	Damera Vyshali	Data visualisation, Project documentation and README
Execution	Sahittya Dutta	Data preprocessing and training, Feature Selection, Requirements and execution instructions
PPT	Haral Anuradha Dnyaneshwar	Data testing and visualisation, Five-slide presentation
Video	Prathmesh Jamdade	Data exploration, evaluation, Project structure and Project video

AI Use Disclosure

OpenAI ChatGPT assisted with drafting the first iteration of README.md and to plot multiple graphs on one png.
Google Search was used to discover ANSI codes and how to print coloured text in terminal.
The project team assumes the responsibility for the code.